

Synthetic Aperture Tilt-Shift from a Handheld Phone Video

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ELEC 549

23 September 2026

1 Creative goal

This project was made with the goal of reproducing tilt-shift photography with only an iPhone. From my understanding, tilt-shift photography creates a miniature appearance by tricking the eye with a very shallow depth of field. I hypothesized that this effect could be replicated with a video shot on iPhone.



Figure 1: The O'Connor lobby, seen from an upper floor. The focus plane is tilted 26° from the optical axis, and the image is the mean of 169 registered frames.

2 How it works

In a normal DSLR, light rays enter and the lens essentially adds all these rays up into a visible image. The limiting factor here is the aperture, which sets how wide a range of positions the rays

from each object can enter through. When you have an object at the focal distance, it appears sharp because all the rays “agree” and center at the same place. On the other hand, objects that are farther and closer have many rays that project to different locations on the focal plane which causes blur. When the aperture is larger, the blur gets larger because the rays from said objects come from a wider spread of positions, so they disagree more.

Now a phone doesn’t have a big lens, and as a result, not a big aperture. As a result of this, it’s impossible to get that depth of field type view. So we can manufacture it. When you record a video with the phone, you can kind of simulate a large aperture by taking in rays from objects at many different perspectives. When I recorded my videos for this project, I would start in the center, and then film in a swirling motion, spiralling outward until I was making large circles with the phone at an arm’s length.

In post processing, you want to layer your images such that the object you want to focus stays clear. Because we moved in a swirling motion, the objects that aren’t focused will naturally blur.

More specifically, the blur comes from uncorrected parallax, and it behaves just like a lens’s “circle of confusion”. When the camera moves sideways by t (as it does during the swirling recording), the image of a point at depth z shifts by about $f t/z$ pixels, where f is the focal length in pixels. The key intuition here is that nearby points shift more than distant ones. If the frames are aligned so that points at the focus depth z_0 overlap, a point at some other depth z is left with a residual displacement $f t |1/z - 1/z_0|$. The translations in the spiral sweep fill a disc of diameter D , so the residual traces a disc of diameter

$$b = D f \left| \frac{1}{z} - \frac{1}{z_0} \right| \quad \text{pixels.}$$

Which is the circle-of-confusion formula for a lens, with the sweep diameter D replacing the aperture diameter. To be clear, this is a simplified version.

The next thing that we do is do something called “plane-induced homography”. This is how we do the tilt-shift effect. We warp each frame in a very particular way, shifting and stretching it by a different amount at every pixel, so that we cancel parallax for every point on one plane. That plane then comes out in focus, even though its points are at many different depths.

$$H = K \left(R + \frac{t n^\top}{d} \right) K^{-1}$$

Acting on a pixel, the product is applied from the right. K^{-1} sends a pixel of the reference frame out along a ray. R rotates that ray by the phone’s rotation between the two frames. The term $t n^\top/d$ contributes the parallax of the plane: it increases with the translation t and decreases with distance to the plane along the ray. The unit normal n and the offset d specify the plane. K maps the corrected ray back to a pixel in the frame being warped. After every frame is warped by its own H , each point of the plane sits where the reference frame recorded it, and the average is sharp across that plane.

3 Capture

I shot the photo of O’Connor from a balcony on the 2nd or 3rd floor. Used my iPhone 13 Pro with the Blackmagic Cam app at 4k, 24fps and with the $1\times$ lens. Then I filmed for 45 seconds in the spiral motion that I mentioned previously, trying to keep the lens pointed in the same direction.

I chose this viewpoint because this is generally how you ought to shoot for the miniature effect. Definitely could have gotten a better shot, but it’s hard to find places with elevation on campus. Additionally, the best miniature shots usually are taken of moving cars, trains, and trucks which would not be possible given the video nature of this technique.

4 Technical steps

You can do this by hand in Photoshop. Load a few dozen frames from the video as layers, then decide which surface should be sharp, the green sofa for example. Translate every layer until that sofa coincides with the sofa in the first layer so that they are stacked. Then average the stack, either with Stack Mode $\rightarrow$ Mean or by giving layer k opacity $1/k$. The sofa is now in the same place on every layer! Note how the table on the floor below is not: each layer laid it down somewhere slightly different, and the mean effect turns the cumulative offsets into a soft smear.

Again, this is the simple version with no tilt-shift. The sharp region stays a slab facing the camera. If you want a tilted plane of focus, it’s a bit more difficult. With Free Transform $\rightarrow$ Distort, mark four points that lie on the desired plane, four corners of the floor tiles for instance, and pull the corners of each layer until those points sit on their counterparts in the first layer. And that’s it, a four-corner distort is a homography. Averaging the registered stack yields an image focused on that tilted plane. Of course, this manual route is impractical since it requires four point correspondences on roughly two hundred layers of images.

Working by eye also fails where the focus surface has no texture, like air. A plane that passes through the air of the atrium, or that lies on a featureless white wall, has no detail at the focus distance to lock onto.

Instead, I wrote a program that automates both stages from camera geometry rather than from dragged points. Every fifth frame is extracted. COLMAP finds thousands of local features in each frame, matches them across neighbouring frames, and triangulates the matches to recover each camera’s position and orientation, the focal length, and a 3D location for every matched feature. The plane is then specified by parameters. Clicking a pixel samples the depth of the nearby triangulated points, and sliders set the tilt while the preview is updated.

5 The final image in numbers

Structure-from-motion recovers shape and leaves absolute scale undetermined, so the distances below are in the reconstruction’s arbitrary units. Pixel blur depends only on the ratio D/z , and the unknown scale cancels in that ratio.

Quantity	Value
Frames	213 extracted, 213 registered by COLMAP, 169 used (aperture 0.8)
Intrinsics	$f = 2658$ px, radial distortion $k = 0.015$, reprojection error 1.27 px
Sweep diameter	10.4 units for the full sweep, 7.8 for the frames used
Scene depth	230–640 units (5th–95th percentile of the sparse points)
Focus plane	tilt 26° , azimuth 351° , depth 284 at the pivot (upper floor by the sofa)

Evaluating $b = D f |1/z - 1/z_0|$ at 4K resolution gives 0 px at the pivot, 1 px at the top of the stairs, 10 px at the reception desk, and 33 px at the table on the lower floor.



Figure 2: Full-resolution crops, with the reference frame on the left and the render on the right. Above, the lower-floor table (predicted blur 33 px) spreads into a soft disc. Below, the sofa, which lies on the focus plane, stays sharp, while a person who crossed the frame during the sweep is left as a faint transparent figure.



Figure 3: The Commons dining hall, one video refocused on two planes. Left: tilt 38.5° along the floor, so near and far tables are sharp while the banners in front and the courtyard beyond are blurred. Right: tilt 63.5° in the opposite sense; the sharp region crosses the mezzanine and the floor is defocused.

6 Validation

The pipeline was checked on rendered imagery before it was applied to the lobby footage. A synthetic city, a textured ground plane viewed from above with upright buildings, was rendered from known camera poses, encoded as a video, and run through the full pipeline. COLMAP recovered $f = 1001$ px against a true focal length of 1000 px and registered all 60 frames. Three clicked points on the ground gave a plane at 54.8° , against a true tilt of 55° . A second test places dots at known depths by direct projection, in code independent of the homography, and measures the width of each rendered dot. The measured widths agree with $D f |1/z - 1/z_0|$ to within 6%. A template-matched shift-and-add of the same frames matches the pose-derived warp to within one grey level.



Figure 4: One synthetic capture refocused at tilts from 0° to 60° , each plane formed in a single pass over the frames.

7 What I learned

The quality of the miniature effect is really decided by how much the depth varies across the scene from your shooting position. This was made clear when I tried to take a picture from the roof of O'Connor. I was far from 30-60-90, and the result was sharp over almost the whole frame, which definitely wasn't what I wanted. The circle-of-confusion formula accounts for it: blur equals $D f |1/z - 1/z_0|$, so it follows the gap between a point's depth and the focus depth, scaled by the sweep diameter relative to distance. A wall seen head-on, or a skyline, has almost no such gap. The frames agree about every object, and no setting of the plane produces blur. The lobby is the contrasting case. One frame contains a column right next to the camera, the furniture on the upper floor, and the study table below, which is about twice as far away as the sofa. With that range the predicted 33 px on the lower floor is clearly visible, and it is still a restrained amount of defocus. Lengthening the sweep, or bringing the subject closer, would increase it. An elevated camera looking down on small objects, the familiar tilt-shift vantage, follows from this scaling. The miniature look is a consequence of the geometry.

Because this is video, a moving person appears as multiple copies, not as a defocused blur. This is because a physical lens integrates every viewpoint at one instant, so a person occupies a single place in all of them and blurs only by being off the plane. The synthetic aperture here is accumulated over 45 seconds. Someone walking through the shot is in a new place in each frame, and absent from most frames, and the average therefore holds a faint transparent image at each place they occupied (Figure 2, bottom right, and the diners in Figure 3).

This method allows us to decide what to focus on in the photo after the fact. A conventional exposure fixes focus when the shutter fires, and the sharp locus is always the flat plane the lens forms, perpendicular to its axis. This clip retains only the viewpoints. The plane brought into focus, its tilt, and the portion of the sweep treated as the aperture are parameters of the average, chosen later. Each warp is built from the camera's recovered position, so the plane is not limited to surfaces that appear in the frames. Following the upper floor at 26° is one setting; a plane that crosses the empty atrium is another. Both, and any plane between them, come out of the same 45-second clip.

What that latitude requires is accurate poses. Place one frame wrongly and the homography misses, spreading the surface it was meant to hold in focus. Figure 1 uses only the inner 80% of the sweep.